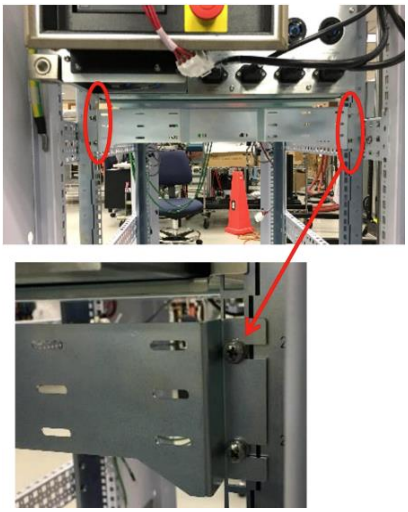
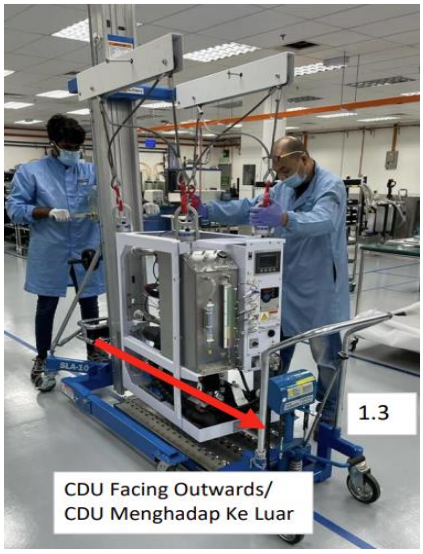
Build Overview:

1 **Ground Mecca Install**
2 **Block Off Panels Install**
3 **SMC Tray Install**
4 **Ethernet Switch Install**
5 **Data Ethernet Switch Install**
6 **Operator Side EMO Switch Install**
7 **Operator THA Interface Assembly**
8 **Operator Interface Panels Install**
9 **CDU Installation**





CHANGE HISTORY

Date	BPM Rev	Change Details	Updated by
	1	New Issue	JOSEPH AMAD

Prepared By:
JOSEPH R. AMAD
Manufacturing Engineer

Checked By:
IVAN NOWELL PANIQUE
Team Lead

Approved By:
SHERWIN ARIOSTE
Operations Manager

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

NO.	PART NUMBER	DESCRIPTION	QTY.
1	357-725-00	Connector	2
2	406-501-05	Screw	2
3	470-501-06	Flat pad	2
4	470-502-07	Spring cushion	2
5	470-933-01	Screw	2
6	472-426-12	Screw	15
7	655-762-00	Blank iron plate	1
8	603-096-00	Nut	56
9	614-766-90	Briquetting	2
10	656-279-10	SMC	1
11	631-020-12	Power Indicator	1
12	631-021-12	Power Indicator	1
13	632-132-00	Baffle	4
14	631-823-00	CDU "Z" Type Braclet	1
15	472-426-12	M5 Screws	4
16	653-315-00	CDU	1
17	633-276-12	MECCA panel	1
18	653-122-00	Blank iron plate	1
19	634-438-00	wire	1
20	634-733-00	AC power panel	1
21	634-736-00	EMO panel	1
22	801-515-02	Connector	1
23	801-541-02	Connector	1
24	801-643-01	EMO Switch	1
25	901-849-01	Ethernet adapter	2
26	479-133-23	label	1
27	634-437-03	Ground wire	1
28	801-513-08	Shield	1
29	654-193-01	Plastic protective cover	2
30	666-796-00	Ethernet switch	1
31	630-893-00	panel	1
32	918-241-15	Power cable	1
33	655-378-00	Ethernet switch	1
34	813-929-00	Tie	1
35	653-553-00	Iron panel	1
36	653-554-00	Iron panel	1
37	653-122-00	Blank iron plate	1
38	657-343-02	Ground wire	2
39	657-343-03	Ground wire	1

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL:

NO.	PART NUMBER	DESCRIPTION	IMAGE
1	N/A	5mm hex bit 13.5 N.m	N/A
2	N/A	8mm Socket Wrench 3.3 N.m	N/A
3	N/A	8mm Socket Wrench 1.6 N.m	N/A
4	N/A	PH2 Phillips Torque Screwdriver 0.7 N.m	N/A
2	N/A	PH2 Phillips Torque Screwdriver 1.6 N.m	N/A
3	N/A	PH2 Phillips Torque Screwdriver 3.3 N.m	N/A
4	N/A	EMO Switch torque wrench 17.7 lbs/ 2 N.m	N/A
2	N/A	Flat Torque Screwdriver 0.7 N.m	N/A
3	N/A	T6 Torx Wrench 0.19 N.m	N/A

METHOD:

Flow chart of operation steps

BPM preparation	See (1. Scope of application)
Preparation Tools	See (2. Tools/Materials/Parameters---1. Tools)
Prepare Materials	See (2. Tools/Materials/Parameters---2. Materials)
Complete Operation	See (3. Operation Steps)
Inventory	See (2. Tools/Materials/Parameters---2. Materials)
Operation Check	See (4. Matters needing attention)
Inventory collection tools	See (2. Tools/Materials/Parameters---1. Tools)

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD: **3. Operation steps**

1.1 Install a grounded MECCA panel (633-276-12) on the right side of the operation surface of the cabinet as shown in the figure, and fix it with the following materials. The tool is a 5mm hexagon socket head. The torque value is 13.5 N·m:

a. 2 pressing blocks (614-766-90).

b. 2 M8 large stud screws (470-933-01).

c. 2 M8 flat washers (470-501-06).

d. 2 M8 spring pads (470-502-07).

Note: the direction of the large diameter G1 and G2.

Note: Please hold the pressure block when installing the screw, otherwise the pressure block will fall.

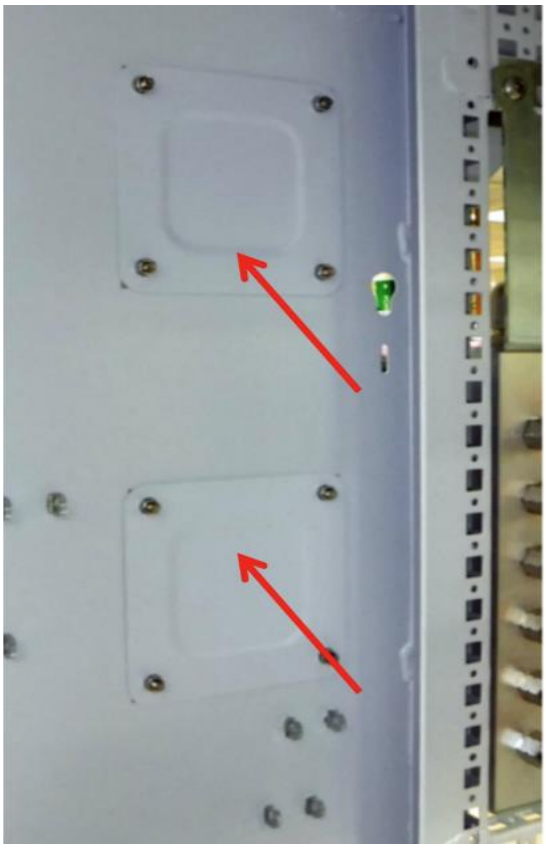
MAN: CERTIFIED OPERATOR, TECHNICIAN

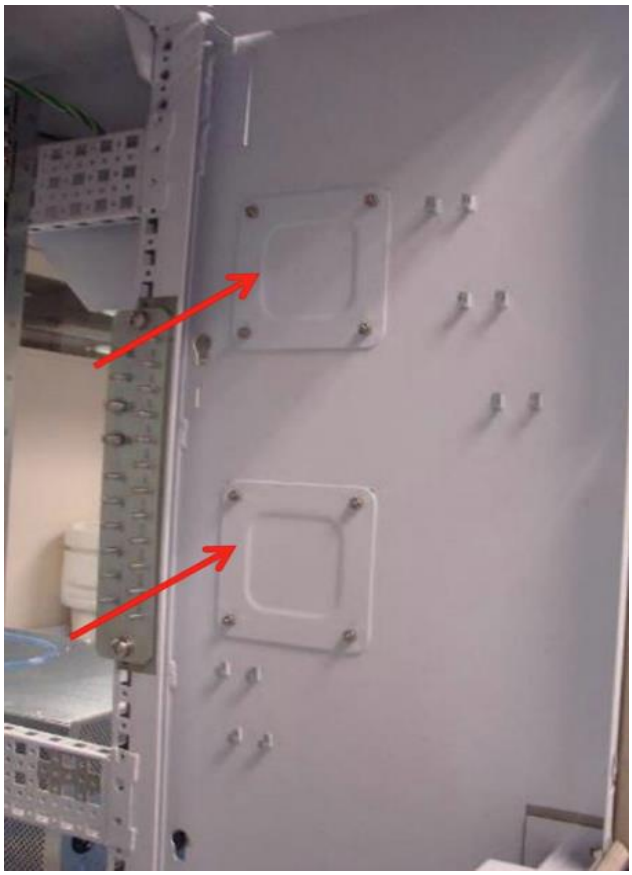
MATERIAL: N/A

MACHINE: N/A

METHOD:

2.1 Install 4 baffles (632-132-00) on the operation surface of the cabinet as shown in the figure, and fix them with 16 M5 hexagon nuts (603-096-00), The tool is an 8mm socket wrench with a torque value of 3.3 N·m.





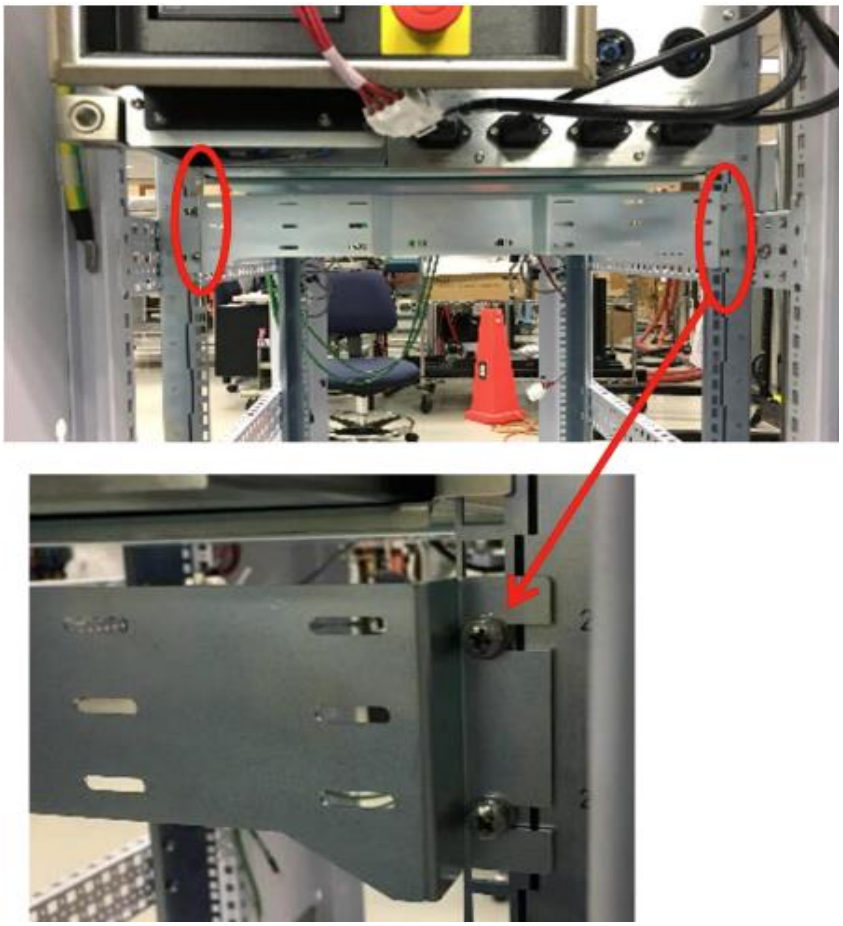
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

2.2 Install a CDU "Z" type bracket (631-823-00) on the maintenance surface of the cabinet, and fix it with 4 M5 screws (472-426-12). The screw positions are 20M and 21M. Don't lock it up yet.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

Install CDU 653-315-00

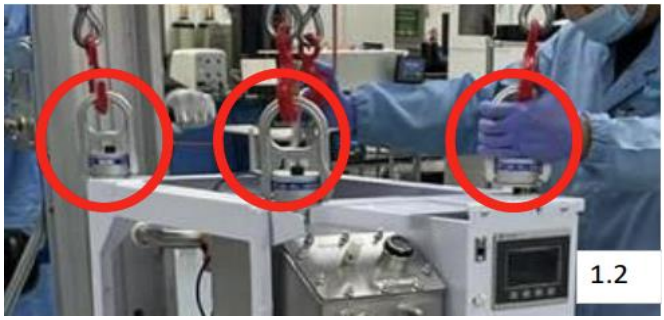
2.3 Use 4 lifting hooks and lock them to the top of the CDU

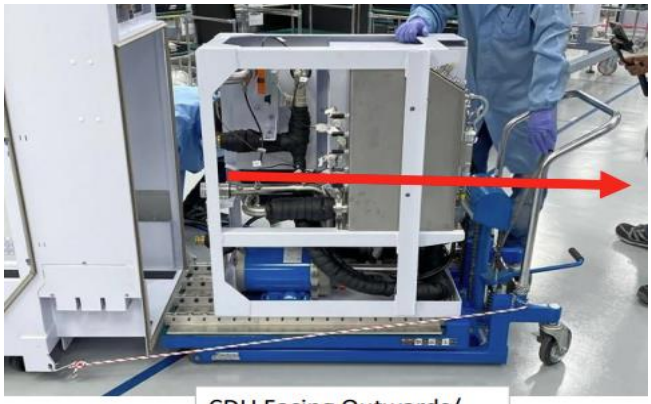
Note: Ensure that bottom of the Screw of the lifting hooks do not come incontact with internal parts

2.4 Make sure that the CDU Panel is facing outwards from the lift Truck then use The Lift Truck and lock the Red Hook to the Lifting hooks, once secured then lift the CDU off the trolley.Pull the trolley to the back and install the Roller Platform onto the trolley.

2.5 Once the Roller Platform is installed, push the trolley and position it underneath the CDU and slowly lower down the CDU.

Note: Ensure that the CDU panel is Facing towards the handle of the CDU trolley. Make sure to hold the CDU so that it does not roll off.





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

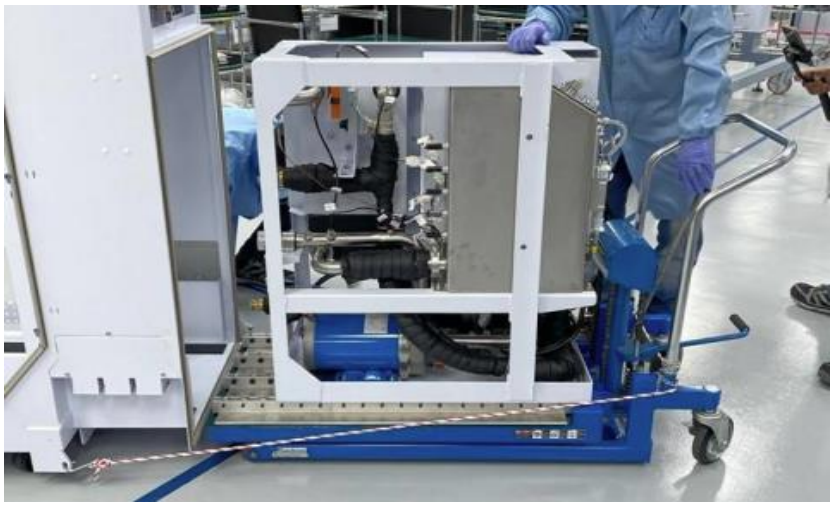
MACHINE: N/A

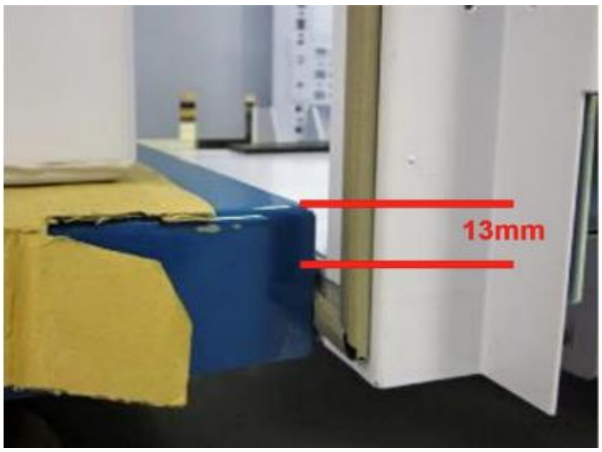
METHOD:

Install CDU 653-315-00

2.6 Place the lift truck with CDU on the operating surface of the cabinet as shown in the figure. The height of the lift truck is about 13mm higher than the cabinet floor.
The water pipe interface of the CDU faces the cabinet.
Note: Ensure that the cables in the cabinet do not interfere with the CDU before installing the CDU.

2.7 Tighten the lifting wheels of the CDU, fix the two fixing holes of the cabinet with a rope as shown in the figure, surround the lift truck, and fix the lift truck to prevent the lift truck from moving back.
A person stands behind and slides the CDU into the cabinet.





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

3.1 Take an SMC (656-279-10) as shown in the figure and remove the cable tie shown in the figure. Take out the black power cord and place it on the left side of the SMC. Re-tie the grounding wire.

3.2 Install the SMC from the operating surface of the cabinet.

3.3 Use 2 screws (472-426-12) to fix the SMC bracket on the cabinet. The tool is a PH2 Phillips torque screwdriver with a torque value of 3.3 N·m.

Cabinet operation surface

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

3.4 The view shown in the figure is a schematic diagram of the SMC on the maintenance side of the cabinet.

Cabinet maintenance surface

3.5 Connect the SMC ground wire along the guide rail to the MECCA G11 interface of the ground panel as shown in the figure, and fix it with a M5 hexagon nut (603-096-00).The tool is an 8mm socket wrench with a torque value of 3.3 N·m.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

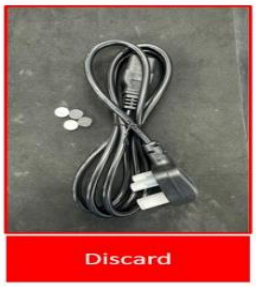
METHOD:

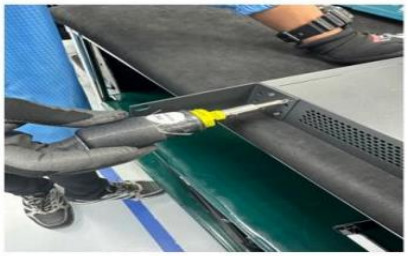
4.1 Take an Ethernet switch (666-796-00).

4.2 Install the bracket on the switch as shown in the figure. Use the built-in fixed material to fix the bracket.
The tool is a PH1 Phillips torque screwdriver, and the torque value is 0.7 N·m.

4.3 Install the switch from the operating surface of the cabinet as shown in the figure. The installation positions are 37L and 37U.
Use 4 M5 screws (472-426-12) to fix. The tool is a PH2 Phillips torque screwdriver with a torque value of 3.3 N·m.
Note: When installing the switch fixing screws, please hold the switch upward.

4.4 Discard the power cord that comes with the switch.





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

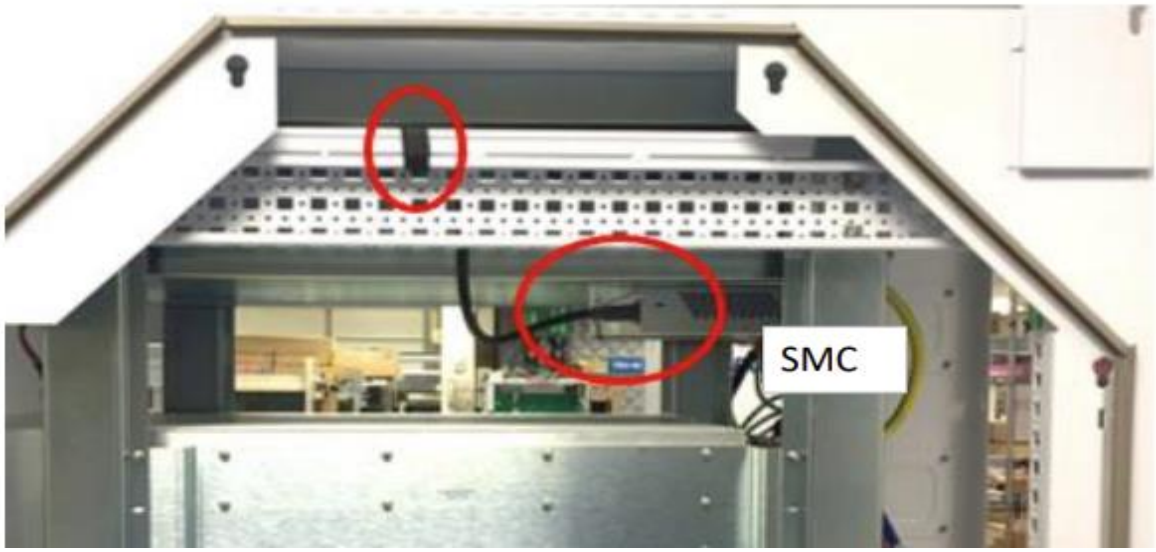
MACHINE: N/A

METHOD:

4.5 Take the AC power cord 918-241-15 and mark the plug end as "ADB J14B"

4.6 Insert the female end of the power cord into the jack on the back of the switch.

4.7 Route the power cord along the left slot of the maintenance surface of the cabinet, and fix it on the bracket with a cable tie 813-929-00.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

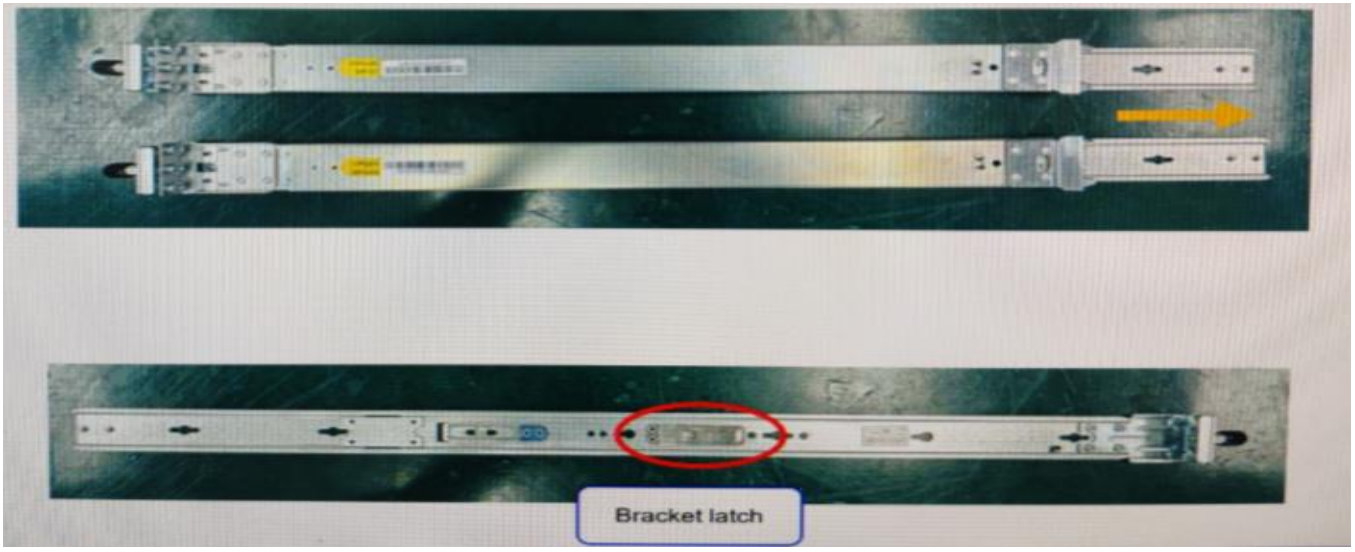
MACHINE: N/A

METHOD:

4.8 Take an Ethernet switch (655-378-30).

4.9 Install the bracket to the **right side of the switch** as shown in the figure, and fix the bracket with the 1 screws that come with it.

4.10 Slide the bracket toward the front of the switch as shown in the figure until the bracket stops on the base, and lock the 1 screws.
The tool is a PH1 Phillips torque screwdriver with a torque value of 0.7 N·m.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

4.11 Install the bracket to the **left side of the switch** as shown in the figure, and fix the bracket with the 1 screws that come with it.

4.12 Slide the bracket toward the front of the switch as shown in the figure until the bracket stops on the base, and lock the 1 screws. The tool is a PH1 Phillips torque screwdriver with a torque value of 0.7 N·m. Check if the bracket is totally installed.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

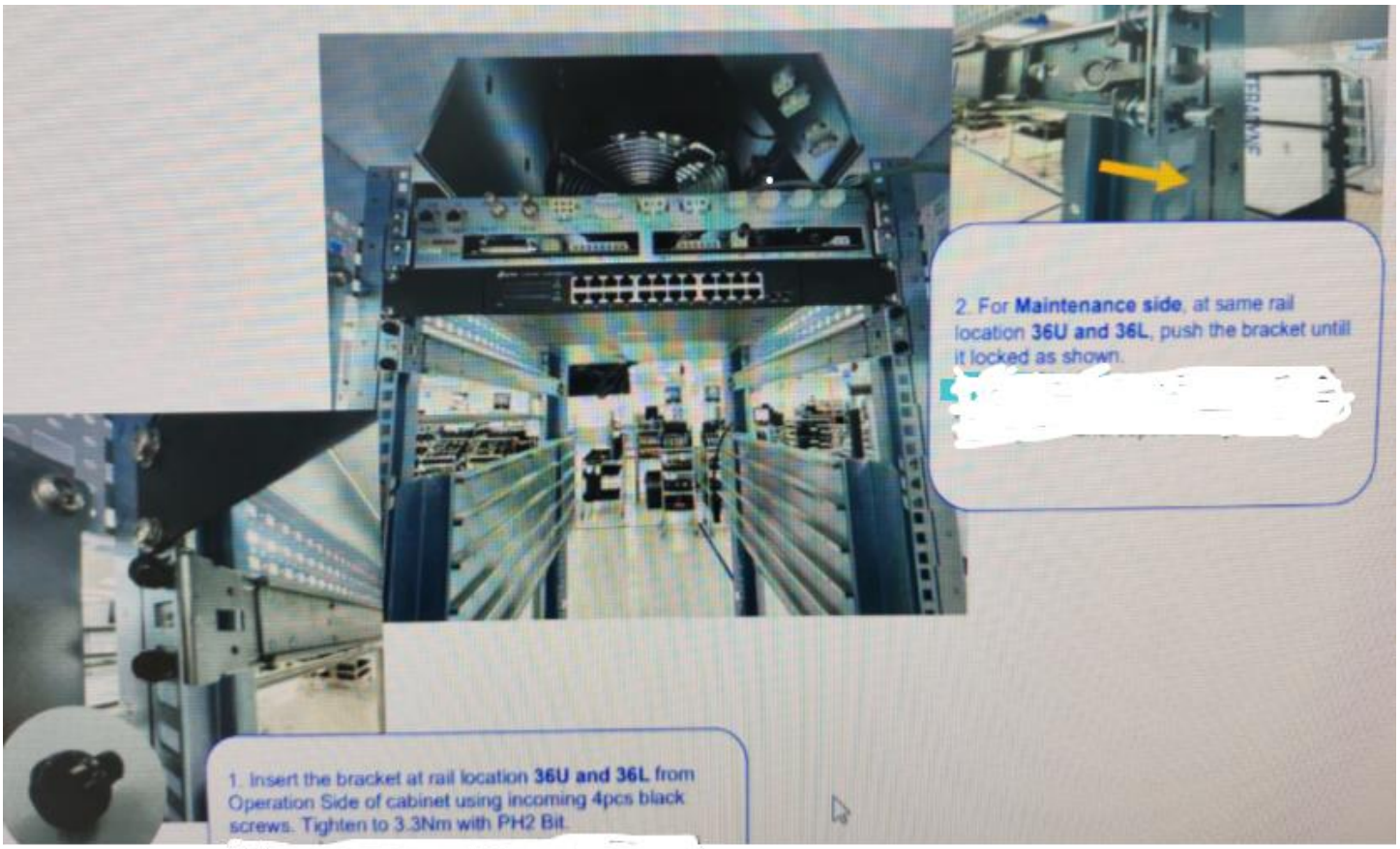
METHOD:

4.13. Prepare the 4pcs black screws that comes with the switch

4.14 . Insert the bracket at rail location 36U and 36L from Operation Side.

4.15. Using PH2 Bit Torque to 3.3Nm

4.16. For Maintenance side, at same rail location 36U and 36L, push the bracket until it locked as shown.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

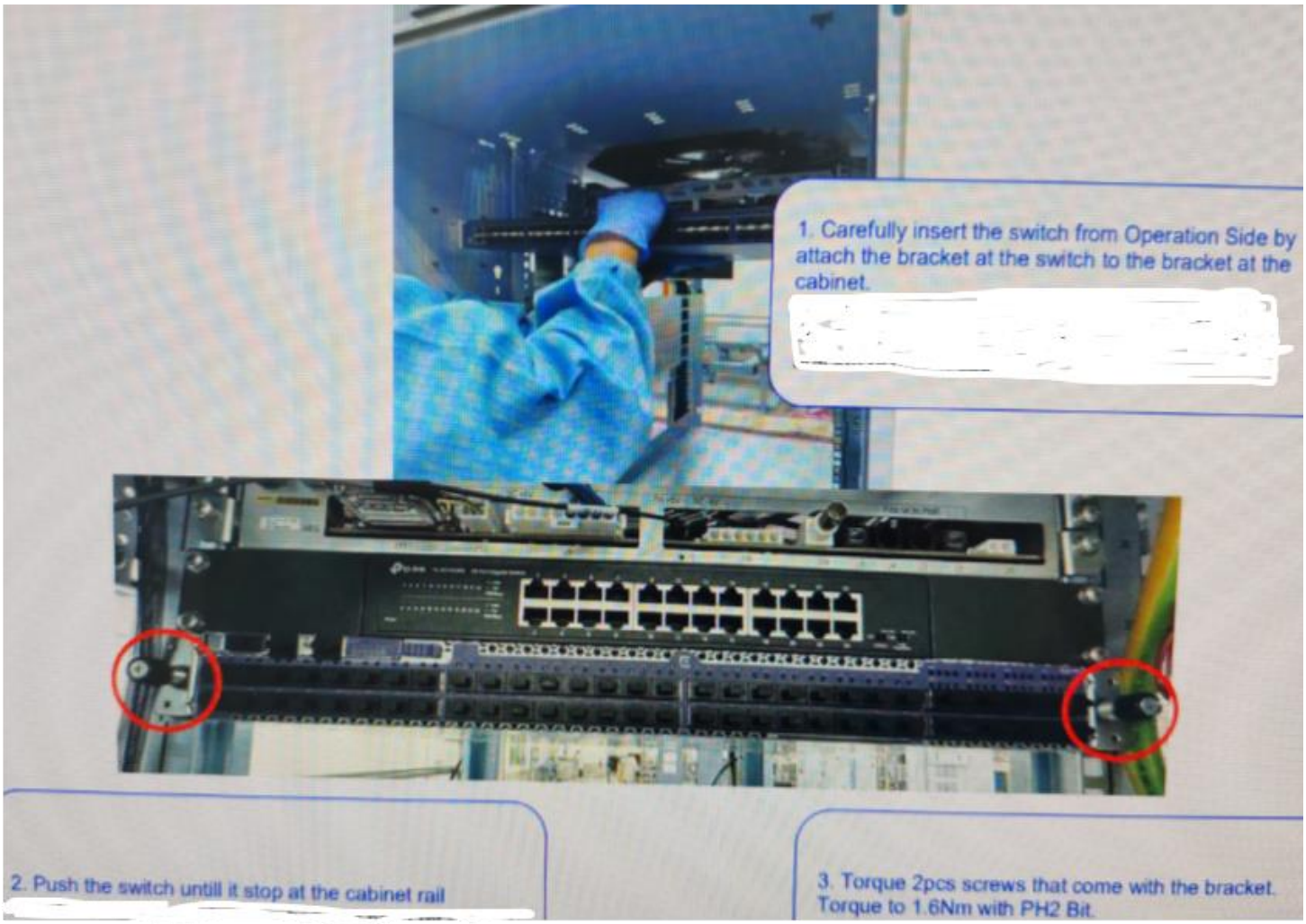
MACHINE: N/A

METHOD:

4.17. Torque 2pcs screws that come with the bracket. Torque to 1.6Nm with PH2 Bit

4.18 Discard the network cable and plug that comes with the switch.

Note: The switch is very heavy. Two people or a suitable lifting device must lift the switch to a suitable location for installation!



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

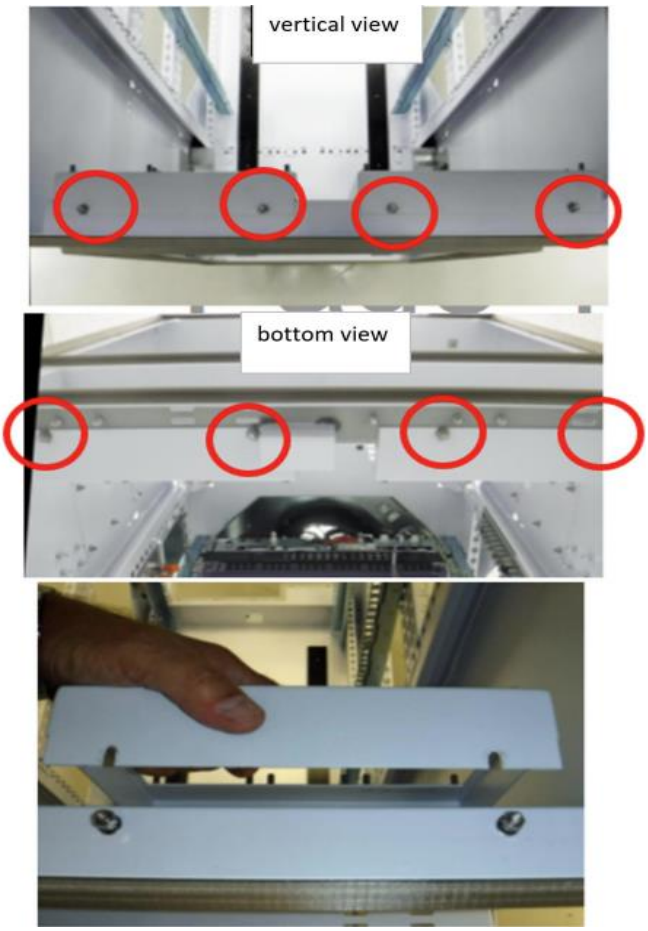
MACHINE: N/A

METHOD:

Cabinet operation panel frame removal---cabinet operation surface

4.19 Loosen 8 M5 fixing nuts, and fix the left operation panel frame to the cabinet-4 on the top and 4 on the bottom.

4.20 Remove the two frames from the cabinet and place them on the operating vehicle for use.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

EMO switch installation

5.1 As shown in the figure, install the EMO warning label 479-133-23 in the center area of the operator interface panel.

MAN: CERTIFIED OPERATOR, TECHNICIAN

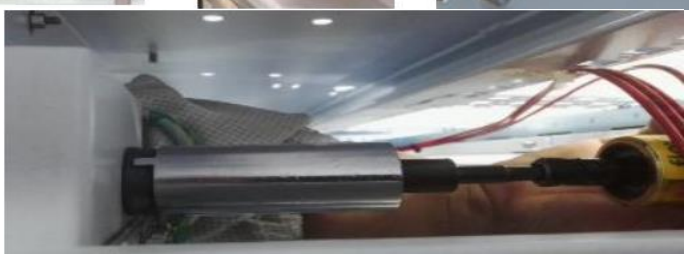
MATERIAL: N/A

MACHINE: N/A

METHOD:

5.2 Install a shield (801-513-08) to the EMO switch (801-643-01). Use the buckle on the top of the shield as shown in the picture
Click on the EMO switch.

5.3 Install the EMO switch (801-643-01) into the hole on the operating surface of the cabinet. Pay attention to the switch direction when installing the switch.
The uppermost text of the switch faces upwards. After installation, use a large plastic nut to lock the switch.
The tool is the icon switch torque wrench, the torque value 17.7 in.lbs. (2 N.m)



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

5.4 Install the second connector (801-541-02) to the main connector (801-515-02) as shown in the figure.

5.5 Follow the steps below to install the wire (634-438-00) on the switch interface as shown in the figure.

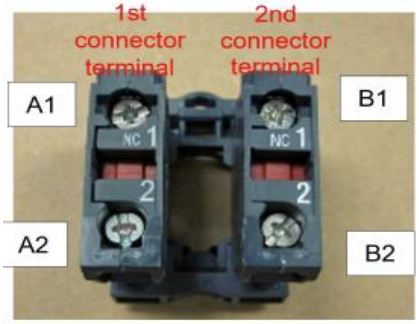
a. A1 is connected to terminal 1 of the 1st connector.

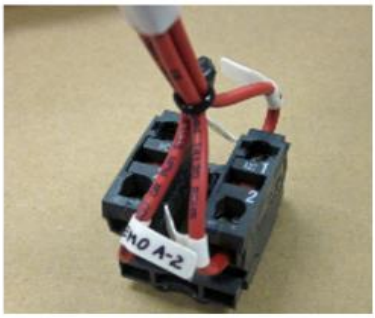
b. B1 is connected to terminal 2 of the 1st connector.

c. A2 is connected to terminal 1 of the 2nd connector.

d. B2 is connected to terminal 2 of the 2nd connector.

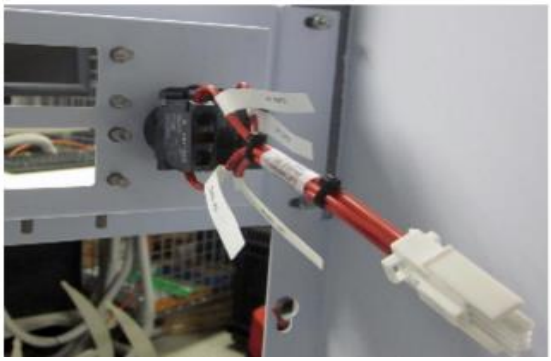
Use a flat-head torque screwdriver with a torque value of 0.7 N·m.

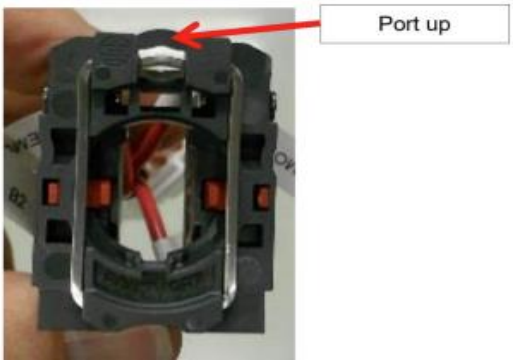




5.6 With the switch port facing upwards, install the EMO connector switch to the back position of the EMO button.

Gently push the button to ensure that the EMO switch can. It is pressed in and rotated out normally without obstacles.





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

6.1 Carefully remove the plastic protective film of the system power indicator (631-021-12) as shown in the figure and stick it to the middle of the panel (653-553-00).

6.2 Install the panel to the center of the Interface panel of the cabinet operation surface as shown in the figure, and fix it with 4 M5 hexagon nuts (603-096-00), The tool is an 8mm socket wrench with a torque value of 1.6 N·m.

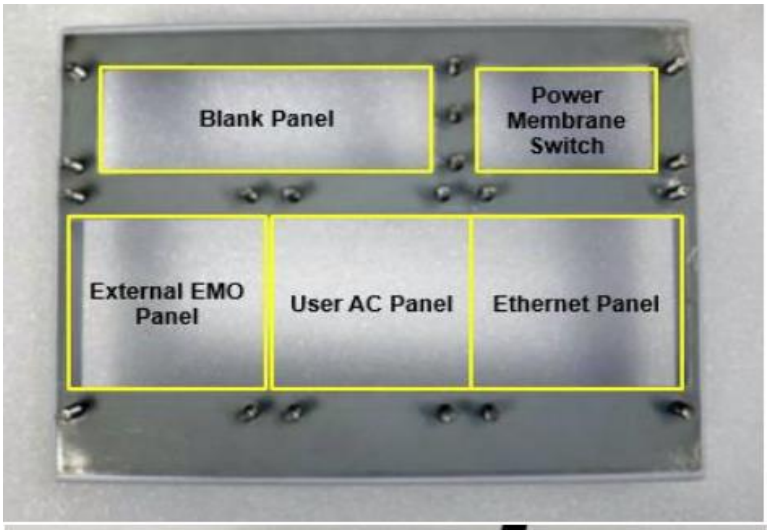
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

7.1 Place one of the previously removed operation panel frames on a flat surface.



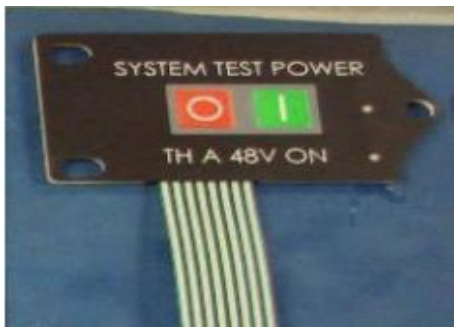
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

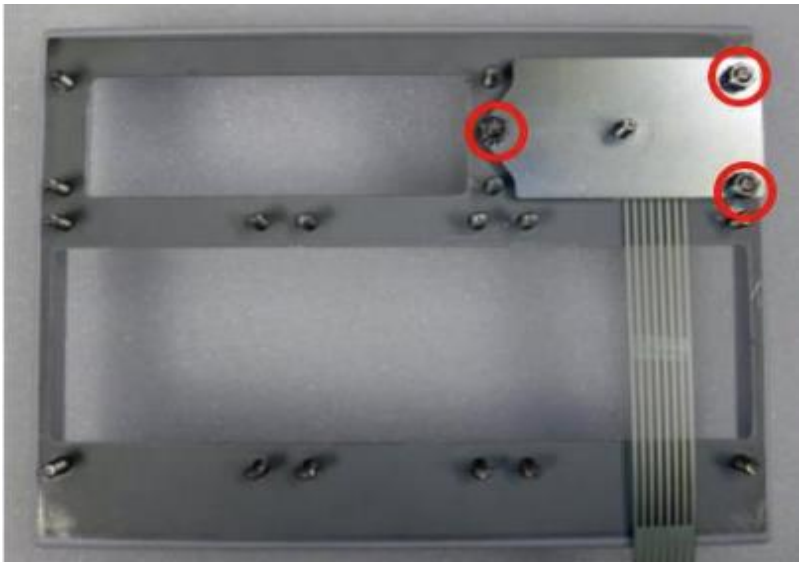
MACHINE: N/A

METHOD:

7.2 Carefully remove the plastic protective film of the other system power indicator (631-020-12) as shown in the figure and stick it to the middle of the panel (653-554-00)



7.3 As shown in the figure, install the panel to the upper leftmost position of the Interface panel on the operation surface of the cabinet, and fix it with three M5 hexagon nuts (603-096-00), The tool is an 8mm socket wrench with a torque value of 1.6 N·m.



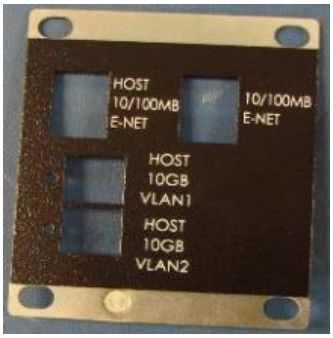
MAN: CERTIFIED OPERATOR, TECHNICIAN

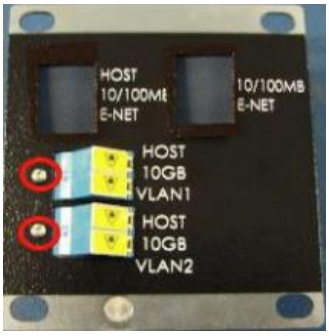
MATERIAL: N/A

MACHINE: N/A

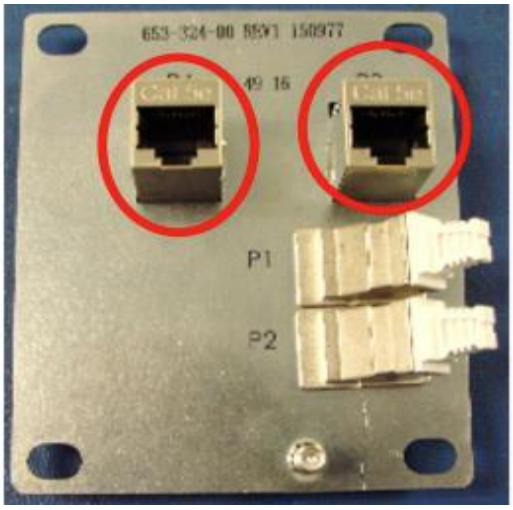
METHOD:

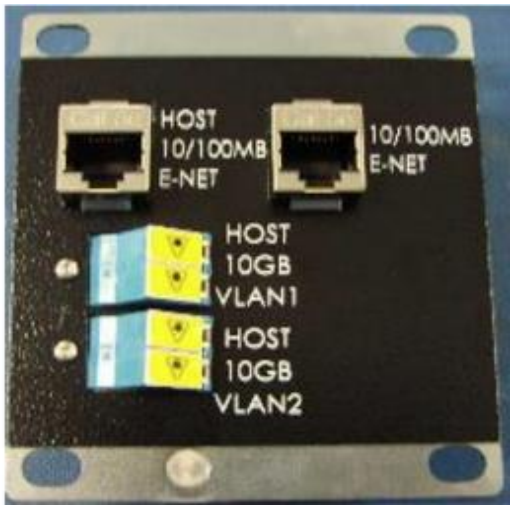
8.1 Install 2 Ethernet adapters (901-849-01) to the "10GB VLAN" on the operation Interface panel (653-324-00) as shown in the figure In the area, use 2 M2 Torx screws (406-501-05) to fix in front of the panel, the tool is a T6 Torx wrench, and the torque value is 0.19 N·m.





8.2 Install two CAT5E connectors (357-725-00) in the remaining space of the panel as shown in the figure. Install from the back of the panel, install the connector diagonally, Then press the connector in.





MAN: CERTIFIED OPERATOR, TECHNICIAN

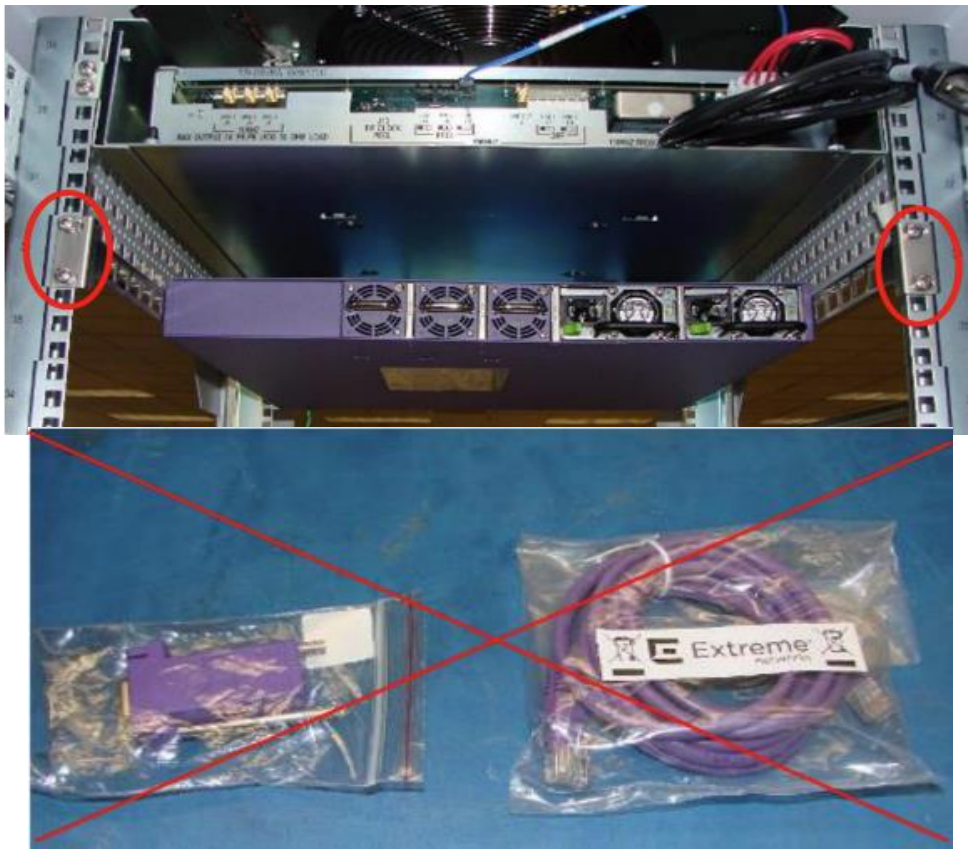
MATERIAL: N/A

MACHINE: N/A

METHOD:

4.17 As shown in the figure, use 4 472-426-12 screws to lock the rear bracket to the rack rail. The tool is a PH2 Phillips torque screwdriver with a torque value of 3.3 N·m.

4.18 Discard the network cable and plug that comes with the switch.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

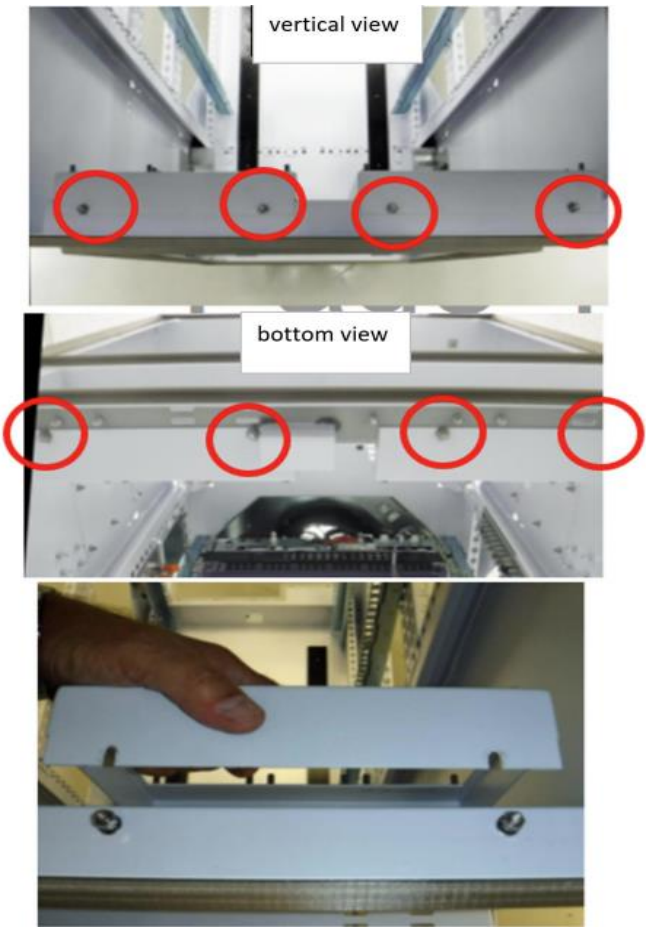
MACHINE: N/A

METHOD:

Cabinet operation panel frame removal---cabinet operation surface

4.19 Loosen 8 M5 fixing nuts, and fix the left operation panel frame to the cabinet-4 on the top and 4 on the bottom.

4.20 Remove the two frames from the cabinet and place them on the operating vehicle for use.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

EMO switch installation

5.1 As shown in the figure, install the EMO warning label 479-133-23 in the center area of the operator interface panel.

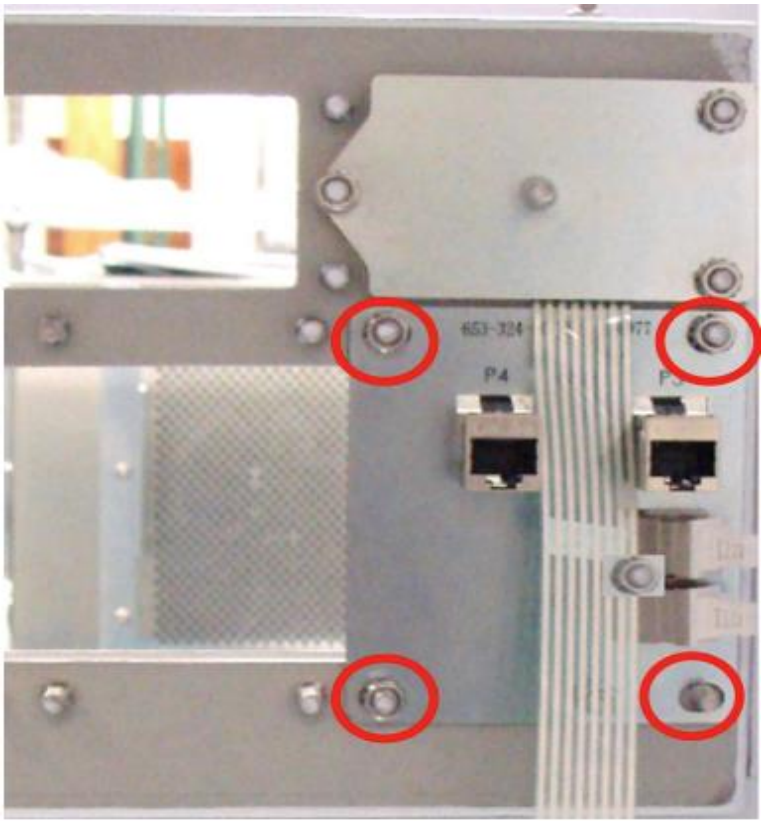
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

8.3 Use 4 M5 nuts 603-096-00 to install the assembled E-Net panel assembly on the bottom right of the INTERFACE panel as shown in the figure, do not lock it first.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

9.1 Use 4 M5 nuts 603-096-00 to install the AC power panel 634-733-00 in the middle of the lower part of the INTERFACE panel as shown in the figure, do not lock it first.

Front View

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

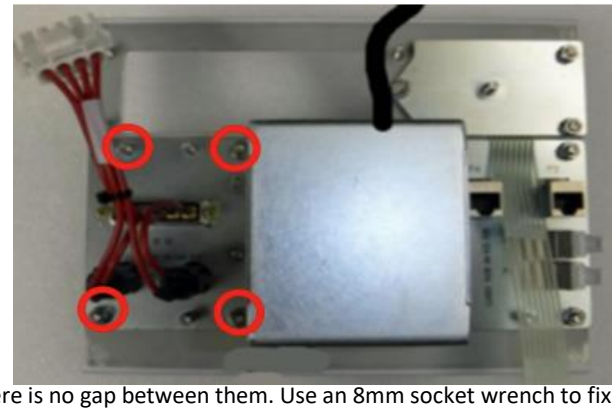
MACHINE: N/A

METHOD:

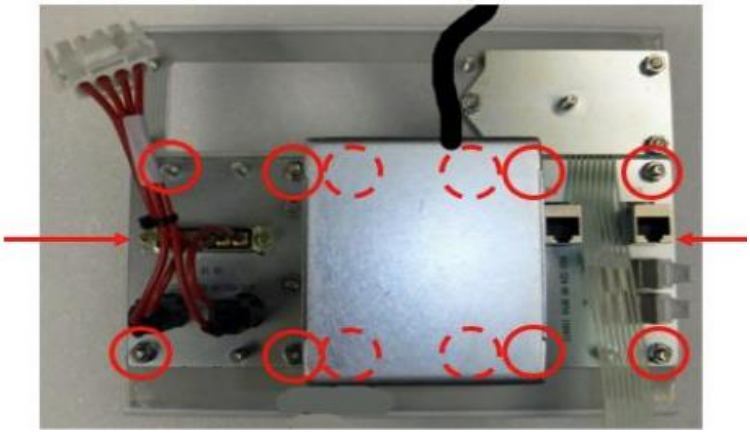
10.1 Install the external EMO panel.

a. Take an external EMO panel (634-736-00).

b. As shown in the figure, install an EMO panel on the right side of the AC power supply panel. Use 4 5mm hexagon nuts (603-096-00) to fix, do not lock it yet.



10.2 Press the 4 panels to ensure that there is no gap between them. Use an 8mm socket wrench to fix the nut with a torque value of 3.3 N·m A total of 12 nuts.



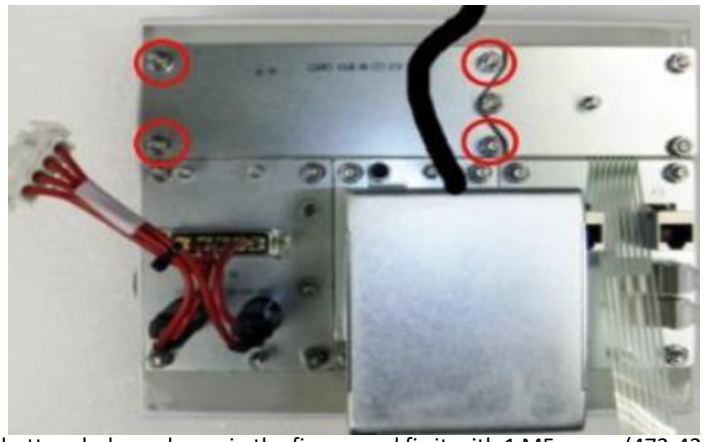
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

10.3 As shown in the figure, install a blank panel (653-122-00) on the upper left of the operation panel and fix it with 4 M5 hexagon nuts (603-096-00), Use an 8mm socket wrench to fix the nut with a torque value of 3.3 N·m.



10.4 Fix the ground wire in the bottom hole as shown in the figure, and fix it with 1 M5 screw (472-426-12), the tool is a PH2 Phillips screwdriver, The torque value is 3.3 N·m.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

Interface panel grounding wire installation

11.1 Take a ground wire 657-343-02, and mark the label as shown: one end is marked as "208V PWR PNL", the other end is "SC_GND_PLATE-G7"
Install the "208v PWR PNL" end on the bottom stud of the AC power module on the Interface panel of the TestHead A,
Fix it with one M5 nut 603-096-00, and fix the nut with an 8mm socket wrench. The torque value is 3.3 N·m.

11.2 Take a ground wire 657-343-02, and mark the label as shown: one end is marked as "EMO PNL", the other end is "SC_GND_PLATE-G8"
Install the "EMO PNL" end on the bottom stud of the EMO panel of the TH A test head Interface panel, Fix it with one M5 nut 603-096-00, and fix the nut with an 8mm socket wrench. The torque value is 3.3 N·m.

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

11.3 Take a ground wire 634-437-03, and install the "MEM INT PNL GND" end on the E-Net panel of the TH A test head Interface panel On the bottom stud, fix it with a M5 nut 603-096-00, and fix the nut with an 8mm socket wrench. The torque value is 3.3 N·m.

11.4 Take a ground wire 657-343-03, and install the “MEM/INTF PNL GND” terminal on the switch panel of the TH A test head Interface panel On the bottom stud, fix it with a M5 nut 603-096-00, and fix the nut with an 8mm socket wrench. The torque value is 3.3 N·m.

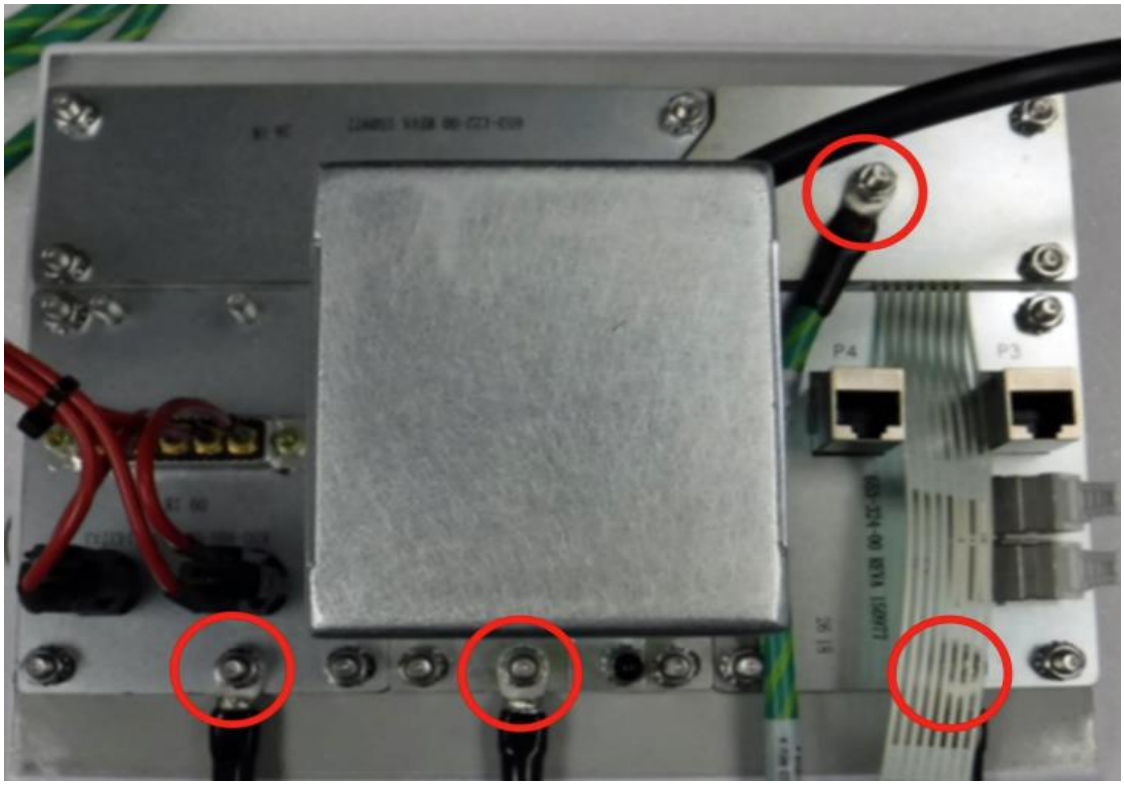
MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

11.5 The figure below is the schematic diagram after all the ground wires of the Interface panel of the TH A test head are connected. Check whether they are installed correctly!



MAN: CERTIFIED OPERATOR, TECHNICIAN

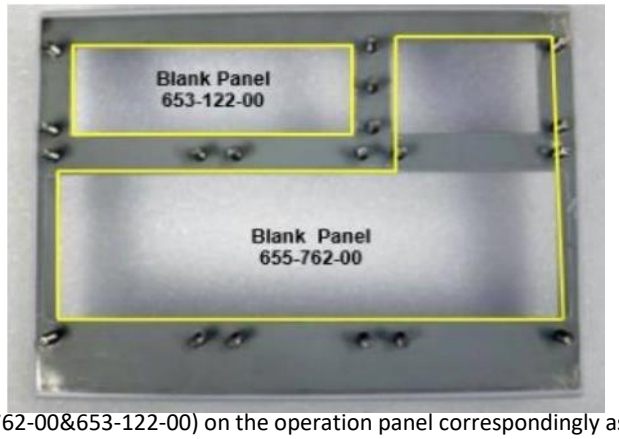
MATERIAL: N/A

MACHINE: N/A

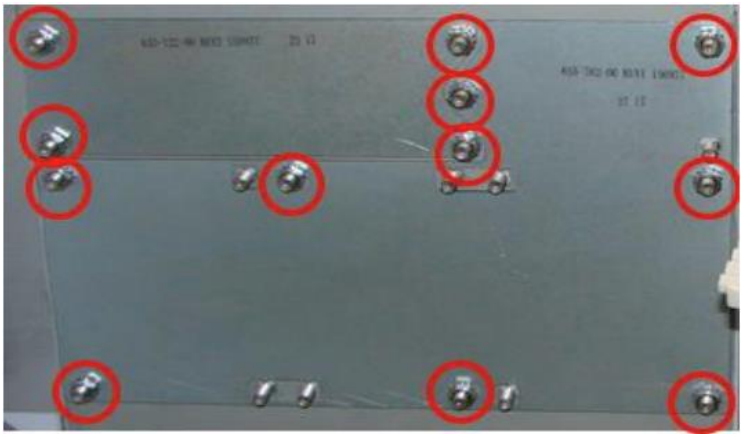
METHOD:

Test Head B Interface panel assembly

12.1 Place one of the previously removed operation panel frames on a flat surface.



12.2 Install the two blank panels (655-762-00&653-122-00) on the operation panel correspondingly as shown in the figure, and use 12 M5 hexagon nuts (603-096-00) Fix, confirm that the two panels are pressed tightly to ensure that there is no gap between them. Use an 8mm socket wrench to fix the nut with a torque value of 3.3 N·m.



MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

13.1 After installation of the two panels of Test Head A and Test Head B are as follows, check whether they are installed correctly! .

Test Head A panel

Test Head B Front view panel

MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

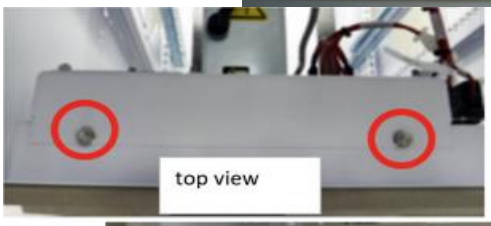
MACHINE: N/A

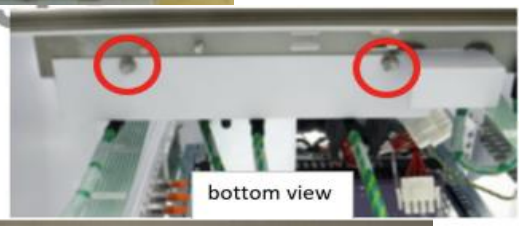
METHOD:

Test Head A panel installation---install from the operating surface

14.1 Install the panel on the left side of the operation surface of the cabinet, paying attention to the position of the flat washer on the top of the panel.

Fasten 4 fixed M5 nuts on the cabinet, and fix the nuts with an 8mm socket wrench. The torque value is 3.3 N·m 2 on the top and 2 on the bottom.





MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

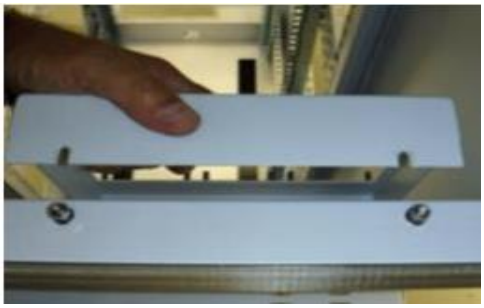
MACHINE: N/A

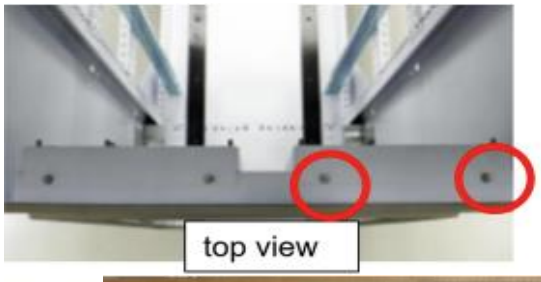
METHOD:

Test Head B panel installation---install from the operating surface

14.2 Install the panel on the right side of the operation surface of the cabinet, paying attention to the position of the flat washer on the top of the panel.

Fasten 4 fixed M5 nuts on the cabinet, and fix the nuts with an 8mm socket wrench. The torque value is 3.3 N·m 2 on the top and 2 on the bottom.



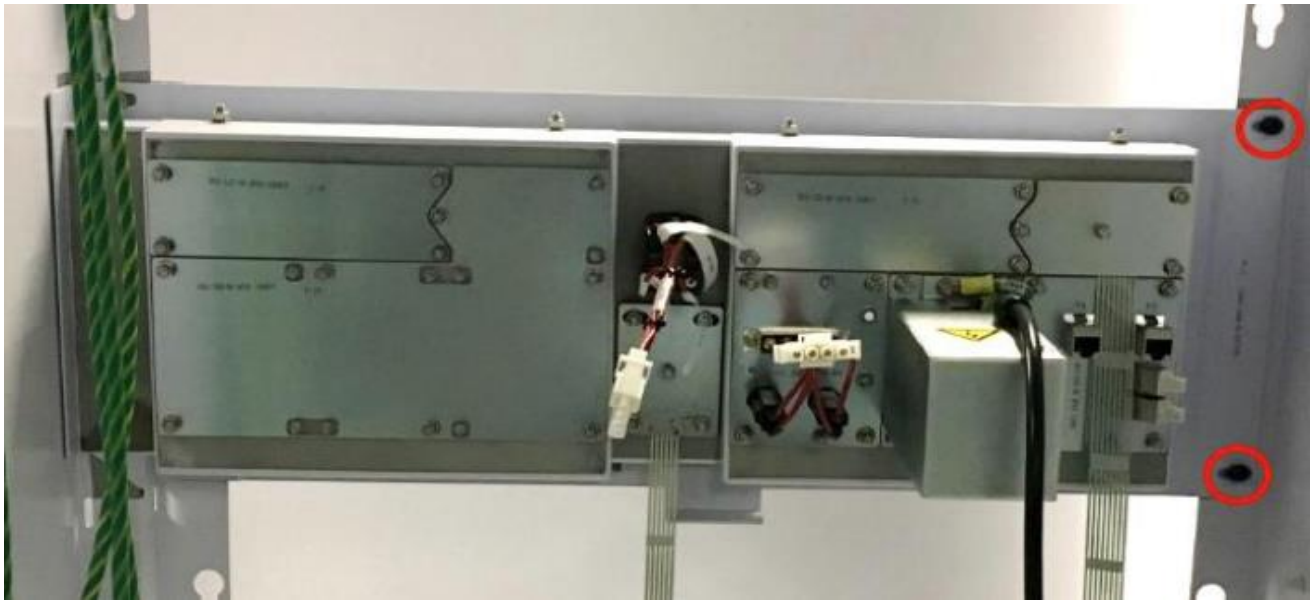


MAN: CERTIFIED OPERATOR, TECHNICIAN

MATERIAL: N/A

MACHINE: N/A

METHOD:

Install the plastic protective cap

15.1 Install two plastic protective caps (654-193-01) in the position shown